Introduction exercises

State assumptions, units, intermediate states, and reasoning. Solutions are in the separate file.

OS01-E01 Enforcing a shared-machine boundary

Alice and Bob share a machine. Bob's program is untrusted.

1. Name three resource boundaries that must be enforced.
2. Identify hardware and software mechanisms for each.
3. When do convenience and isolation conflict?

OS01-E02 Events, modes, and context

A timer interrupts a thread already in kernel mode. Separately, user code calls a cache-hit read. DMA can also discover completion by polling.

1. Classify the sources and synchrony of the events.
2. How many mode/process switches are required in each execution scenario?
3. Can DMA complete correctly without an interrupt?

OS01-E03 Hash counterexamples and lifetime

Use h(k)=k mod4, tail-insert chaining, and head-first lookup. Insert5,9,13. Separately, an updater unlinks node N while a reader still holds N.

1. How many comparisons find 13 and reject 17? What is the worst-case bound?
2. Which additional operations and guarantees can a balanced tree provide?
3. Why is immediate free after unlink unsafe?

OS01-E04 Overlapping two jobs

One CPU, A=CPU 2→I/O6→CPU 2, B=CPU 4, both ready at0, free switching, and overlapping CPU/I/O.

1. Compute completion time and CPU utilization if A finishes before B starts.
2. Give the timeline and results if B runs while A blocks.
3. When does this improvement model cease to apply?

OS01-E05 Cache and NUMA denominators

L1 hits cost1 cycle, hit probability0.9, and misses add50 cycles. Separately, local memory costs100 ns and remote180 ns, with80% local and no overlap.

1. Compute AMAT, and recompute if50 means total miss cost.
2. Compute NUMA means for80% and20% local access.
3. Can their ratio directly give whole-program speedup?

OS01-E06 Equal VA, unequal data

Pages are4 KiB. P maps VPN 1→PFN 7 RW; Q maps VPN 1→PFN 9 RO.

1. Decompose VA0x1234 and compute both PAs.
2. What happens to writes by P and Q?
3. Can frame 9 be freed after Q releases it if another mapping still references it?

OS01-E07 Timer and fairness premises

RR has n continuously ready jobs, quantum q, and switching cost c. A kernel handler may fail to terminate.

1. Give a conservative rotation-wait bound.
2. Compute CPU-only rotation overhead for q 4 ms and c0.1 ms.
3. Does a timer alone ensure progress for every ready job?

OS01-E08 Lost update and BST range

Counter starts0. T1 and T2 each load, add1, and store. A tree has root 10, left5, right20, and12 as5's right child.

1. Give a valid interleaving ending at1.
2. Why is this not a BST, and how does search12 fail?
3. How many bytes does a bitmap for2^20blocks require?

OS01-E09 Deadlines and platform boundaries

A(C1, T=D4), B(C2, T=D6), one preemptible CPU, no overhead or blocking.

1. Give a feasible0–12 schedule and all deadlines.
2. Does utilization below1 guarantee deadlines without naming a scheduler?
3. Compare VM and conventional-container kernel boundaries.

OS01-E10 API, version, and source availability

A service offers downloadable source without stated redistribution/modification rights. A program claims crash durability after one successful write.

1. Does readable source establish open source?
2. Can one observation prove every API guarantee?
3. Which contracts are missing between write completion and durability?

OS01-E11 Print jobs and two completion times

One printer handles one page per second without interleaving jobs, using FCFS. A's six pages arrive completely at the spool at t=0 and B's two at t=1. Ignore transfer time and start the printer immediately. List queue and printer intervals, B's waiting time and output-completion time. Does successful submission at t=1 prove B has been printed or will survive a power failure? What do the spool and a buffer each do?

OS01-E12 Concurrency, parallelism, and shared state

P and Q each need two seconds of CPU. Scenario A uses one core: P0–1, Q1–2, P2–3, Q3–4. Scenario B uses two cores: P and Q each occupy one core during 0–2. Compare concurrency, parallelism, completion time, and total CPU work. If both update a shared counter, does parallel execution itself guarantee correct updates?

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 1. Open Source Initiative: Open Source Definition.

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https://github.com/xinhi6716/computer-science-self-study